Source interfaces for characteristic-two orthogonal markings

Human Compression project: L031 corrected recovery

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This paper supplies the source-specific interfaces and auxiliary qualifications for the complete corrected recovery of Brooksbank–Ferrara–Leemans [BFL]. The full companion papers are required: *Endpoint variation in orthogonal symmetry markings*, *The four displayed symplectic generators*, the two binary boundary/exclusion papers, the centralizer rank-bound paper, and the previously accepted *Projective involutions and support spaces in small dimension*. None is replaced by a reference to its conclusions alone. The classical results identified below remain external inputs.

1 Forms and the exact support boundary

We use the source's definition of an orthogonal space: the polar radical has dimension at most one. For a one-dimensional radical $\langle z\rangle$ choose a nondegenerate complement M. If $Q(z) = 0$, every isometry is uniquely of the form

$$z \longmapsto cz, \qquad x \longmapsto hx + f(x)z \quad (x \in M), \qquad c \in F^\times, h \in O(M), f \in M^*.$$

This proves the split description (2.3), with additive radical M^*, identified with M through the nondegenerate pairing. If $Q(z) \neq 0$, an isometry fixes z. Conversely each symplectic map h on M has a unique lift $x \mapsto hx + f(x)z$, where $f(x)^2Q(z) = Q(x) + Q(hx)$. The right side is a quadratic function with zero polar form; over the perfect field it is the square of a unique linear function times $Q(z)$. This proves (2.4) directly.

In a nondegenerate plane, hyperbolic versus asingular means that the binary quadratic polynomial has respectively two or no projective roots. If the polar form is zero, the quadratic form is a squared linear function and has either one singular point or is identically zero. This gives all four source line types. For nonradical nonsingular x, the formula $s_x(v) = v + B(v, x)x/Q(x)$ is a nonidentity involution with support $\langle x\rangle$. Distinct such symmetries commute exactly when their root vectors are perpendicular, as follows by multiplying their rank-one differences from the identity. Scaling the root by a *nonzero* scalar leaves the symmetry unchanged. A nonsingular *radical* root instead gives the identity; the source's generic involution/support language must be read with this qualification.

Theorem 1 (The larger-field intersection statement)**.** *For $q \geq 4$, the conclusion of [BFL, Proposition 3.3] holds in every orthogonal ambient space allowed there, including a singular one-dimensional radical.*

Proof. For a nonsingular ambient space this is exactly the support characterization in the endpoint companion, applied to the two subspaces and their intersection. If its radical $\langle z\rangle$ is singular, adjoin a vector w with $B(z, w) = 1$ and $Q(w) = 0$, perpendicular to a chosen nondegenerate complement of z. The larger space is nondegenerate. All its symmetries rooted in the old space preserve that space. Restriction of their whole group is faithful: an element supported in the old space and fixing it pointwise has the form $x \mapsto x + cB(x, z)z$; quadratic preservation and $Q(z) = 0$ force $c = 0$. The proved ambient intersection equality therefore restricts to the asserted one. □

Theorem 2 (The unrestricted binary proposition is false)**.** *The same Proposition 3.3, literally stated without excluding $q = 2$, has a counterexample with* $\dim V = 8$, $\dim W_i = 6$ *and* $\dim(W_1 \cap W_2) = 5$.

Proof. Let $M = M_2(\mathbf{F}_2)$ with $Q(X) = \det X$, the plus quadratic four-space. Its nine nonzero singular vectors are the rank-one matrices uv^{T}, a 3 by 3 grid indexed by the nonzero vectors $u, v \in \mathbf{F}_2^2$. Its maximal totally singular planes are the three row fibers and three column fibers of this grid: if two independent rank-one matrices have independent left and right factors, their sum has rank two. Thus an isometry permutes these two families, possibly interchanging them. The product of the signs of its two induced S_3 permutations is a well-defined homomorphism $\chi : O(M) \to \{1, -1\}$, also when the families are interchanged.

Every nonsingular matrix A has determinant one. The polar identity for the determinant gives the symmetry formula

$$s_A(X) = A \operatorname{adj}(X) A.$$

Left and right multiplication by A have the same ruling sign. Moreover $\operatorname{adj}(X) = JX^{\mathsf{T}}J$, where $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; the two J signs cancel and transposition only swaps the families. Consequently $\chi(s_A) = 1$ for every symmetry. Left multiplication $g : X \mapsto JX$, however, has $\chi(g) = -1$. In particular $g \notin \Sigma_M$.

Now take $V = M \perp H_1 \perp H_2$, where H_i have singular hyperbolic bases e_i, f_i with $B(e_i, f_i) = 1$. Put

$$U = M + \langle e_1 \rangle, \quad W_1 = M + \langle e_1, f_1 \rangle, \quad W_2 = M + \langle e_1, f_1 + e_2 \rangle.$$

Both W_i are nondegenerate six-spaces and their intersection is U, whose radical is the singular line $\langle e_1 \rangle$. Extend g by the identity on $H_1 \perp H_2$. It is supported in M and preserves both W_i and their perpendicular complements. Each full six-dimensional orthogonal group is generated by symmetries; this is the explicitly retained classical generation result [Taylor, Corollary 11.42], whose sole exception is $O^+(4, 2)$. Therefore $g \in \Sigma_{W_1} \cap \Sigma_{W_2}$. Every symmetry rooted in U restricts on $U/\langle e_1 \rangle = M$ to a symmetry of M. Since g induces the already excluded map on this quotient, $g \notin \Sigma_U$. This proves the literal failure. No finite computation is a premise. □

The positive construction uses only the proved $q \geq 4$ statement. The two separate binary papers give the corrected existential dimension boundary; they do not use Proposition 3.3.

2 Three-spaces and full-order interval generation

The intrinsic statement [BFL, Lemma 3.4] follows as well. For nonsingular radical, $\Sigma_V = \mathrm{SL}_2(q)$ by (2.4), and the asingular plane gives its maximal $D_{2(q+1)}$. For singular radical, writing $V = L \perp \langle z \rangle$ identifies Σ_V with $L^+ \rtimes O^-(L)$. All translations follow from $s_v s_{v+az}$ as in the companion. The cyclic group of order $q + 1$ is irreducible on L even over $\mathbf{F}_2$; hence any subgroup properly containing the complement contains all translations. This proves maximality for arbitrary elements, not only a chosen third symmetry. When $q = 2$ and the radical is nonsingular, the two groups are both S_3, as stated in Remark 3.5.

For a hyperbolic plane and $q \geq 8$, replace the torus by $\operatorname{diag}(a, a^{-1})$, a primitive in $\mathbf{F}_q^\times$, and include the interchange of its two axes. The minimal polynomials of a and a^{-1} over $\mathbf{F}_2$ are different: equality would give $2^j \equiv -1 \pmod{2^k - 1}$ for some $0 \leq j < k$, impossible when $k \geq 3$. They each have degree k, so the two torus constituents are irreducible, nonisomorphic, and interchanged by the involution. The full dihedral module is therefore irreducible over $\mathbf{F}_2$. The singular-radical argument just given works again. For nonsingular radical retain the split torus-normalizer maximality in $\mathrm{SL}_2(q)$. This proves the $q \geq 8$ hyperbolic assertion in Remark 3.5. For $q = 4$, take $a \in \mathbf{F}_4$ of order three. The orbit of $(1, 1)$ under $\langle \operatorname{diag}(a, a^{-1}), \mathrm{swap} \rangle$ consists of the three nonzero vectors of a proper two-dimensional $\mathbf{F}_2$-subspace. Its translations with the complement form the proper intermediate group asserted in that remark.

The rank-one maximality input here is the classical Dickson subgroup classification: in even characteristic both full torus normalizers $D_{2(q\pm1)}$ are maximal for $q \geq 4$. An exact locally available formulation is Proposition 2.3, page 5, of Leemans–Moerenhout–O'Reilly-Regueiro, *Projective linear groups as automorphism groups of chiral polytopes*, citing Bray–Holt–Roney-Dougal and Dickson. It is retained classification, not proved by these constructions.

Now take the source path with every adjacent coefficient giving order $q+1$. The endpoint companion proves generation of every interval of length at least three; length two is its full dihedral plane and length one is immediate. If the full ambient path has singular radical, adjoin one more nonzero end coefficient to make an even nondegenerate path. The same interval proofs hold in that ambient space, and their restriction to the original space is faithful by the preceding embedding argument. This proves all of Lemma 5.3, and also its natural length-d extension used in the source proof. The full IP induction follows the same way; it identifies the generated *symmetry group* when the radical is singular, not the additional radical scalings in (2.3).

For $q \geq 8$ the mixed $q-1, q+1$ assertion of Remark 5.4 follows by the same induction using the hyperbolic version just proved. At each step use the first marked two-space L and the suffix hyperplane W. Their intersection contains the nonsingular second basis root. Thus the intermediate two-space has at most two singular points; exclude also the single point in L and the possible radical point to obtain the third symmetry needed in the ambient three-space argument. In the nonsingular-radical case its central kernel is recovered by the same Dickson-character and perfect-SL_2 argument as in the companion. This proves the asserted generation without silently extending an intrinsic three-space calculation to the ambient group.

For $q = 4$, a bond of order three forces its normalized coefficient to be one. A length-b all-three path is therefore the scalar extension of the binary path, whose group is S_{b+1} by the paid binary companion. Thus the source's symmetric-group subscripts are one too small: $t+1$ consecutive generators give S_{t+2}, and d binary path generators give S_{d+1}.

3 Radicals and the Arf prescription

For the path $Q = \sum x_i^2 + \sum \alpha_i x_i x_{i+1}$, successively solving its tridiagonal polar equations proves Lemma 4.2 in its full stated range. The even determinant is $(\alpha_1\alpha_3\cdots\alpha_{d-1})^2$. In odd dimension $2m+1$ the radical vector has alternate entries

$$z = (1, 0, \beta_1, 0, \ldots, 0, \beta_m), \qquad \beta_j = \prod_{i=1}^{j} \alpha_{2i-1}/\alpha_{2i},$$

and $Q(z) = 1 + \sum \beta_i^2$.

In even dimension set $e_1 = v_1$ and, recursively,

$$e_i = v_{2i-1} + (\alpha_{2i-2}/\alpha_{2i-3})e_{i-1}, \qquad f_i = v_{2i}/\alpha_{2i-1}.$$

The odd basis vectors are mutually perpendicular, as are the even ones. Direct cancellation gives $B(e_i, f_j) = \delta_{ij}$. Expanding the recursion is exactly (4.5); quadratic evaluation gives $Q(e_i) = 1 + (\alpha_{2i-2}/\alpha_{2i-3})^2 Q(e_{i-1})$. Thus the Arf trace is $\mathrm{Tr}\sum_i Q(e_i)/\alpha_{2i-1}^2$. For $m \geq 2$, equal endpoint coefficients λ and equal interior coefficients μ, both full-order parameters, this is

$$\binom{m-1}{2} \mathrm{Tr}(\mu^{-1}) + \mathrm{Tr}(\mu/\lambda^2).$$

Since $\mathrm{Tr}(\lambda^{-1}) = \mathrm{Tr}(\mu^{-1}) = 1$, this equals the trace of the printed expression (5.6). Passing between inverse squares and inverses is legitimate modulo $\{x^2 + x : x \in F\}$. The primitive norm-one eigenvalues, paired with their inverses, give exactly $\varphi(q+1)/2$ full-order parameters, so their existence and the source parameter sets are explicit.

The subsequent source instruction to choose a primitive μ with an additional trace constraint is not used as an existence premise here. The complete endpoint-variation proof replaces that instruction by a surjective trace calculation with freely variable nonzero endpoints. Both Witt types and every stated dimension are obtained. In dimension two the two dihedral groups are treated directly, rather than substituting into a formula with an unused interior coefficient. This is a corrected proof of the full principal type-existence conclusion, not a claim that the source's particular two-primitive-parameter prescription has been certified.

4 Complete low-dimensional principal spectrum

The accepted whole paper *Projective involutions and support spaces in small dimension*, from the L011 recovery, supplies the even-characteristic exclusions for $\mathrm{PSL}_3(q)$, $\mathrm{PSU}_3(q)$, $\mathrm{PSL}_4(q)$ and $\mathrm{PSU}_4(q)$. Its canonical projective lifts and both six-dimensional quadratic types are explicit. It retains the classical natural-module facts, Dickson invariant and Klein identifications; none of its exclusions uses the L031 theorem being recovered. The entire proof paper is an acquired dependency here, not a circular citation to L011's classification application.

For $\mathrm{SL}_2(q)$ in characteristic two, the centralizer of a nonidentity involution is the abelian upper-unipotent group in an appropriate basis. In a connected string of length four, its first generator centralizes the noncommuting last pair, impossible. This also excludes the natural rank-one classical identifications appearing in dimensions three and four, including $\Omega_4^-(q) \cong \mathrm{SL}_2(q^2)$. The plus four-dimensional derived orthogonal group is a product of two rank-one factors and is not quasisimple.

For the remaining quasisimple group $\mathrm{Sp}(4,q)$, $q \geq 4$, the endpoint companion gives rank five, the literal four-matrix companion gives rank four, and the complete root-authored centralizer companion excludes every connected string of length six or more. Simplicity ensures connectedness for a full-group string C-marking. These prove both directions of Theorem 6.1 for its stated ranks $r \geq 4$, with no retained maximal-subgroup rank assertion and no finite certificate.

For completeness its small-field aside also has an elementary check. The full $\mathrm{Sp}(4,2) \cong S_6$ is not quasisimple. In its derived group A_6, every involution is a double transposition and has centralizer D_8. Thus a string C-marking of rank at least five would place its last three of the first five generators in a dihedral centralizer. An irreducible rank-three C-marking cannot generate a dihedral subgroup: its commuting endpoints differ by a central half-turn, contradicting the rank-three intersection property. The rank-four case is equally direct. Normalize the first generator to $(12)(34)$. A distinct commuting last generator is, up to its centralizer, either $(13)(24)$ or $(12)(56)$. In the first case the two middle generators, required not to commute with their neighbours, belong respectively to

$$\{(13)(56),(24)(56)\}, \quad \{(12)(56),(34)(56)\}.$$

All four marks preserve $\{1,2,3,4\} \mid \{5,6\}$. In the second case their respective possibilities are

$$\{(15)(26),(16)(25)\}, \quad \{(13)(24),(14)(23)\};$$

all four preserve the three pairs $\{1,2\},\{3,4\},\{5,6\}$. Neither case generates A_6. The lists follow simply by commuting a double transposition with each normalized endpoint, and exhaust the four other involutions of each D_8 centralizer.

What is not a recovered question. The odd-prime modular-reduction Theorem 1.3 is attributed prior work and is not a new proof or a premise of the characteristic-two construction. The regular-polytope/string-C-group correspondence and the end-parabolic criterion are retained from McMullen–Schulte. The historical availability of Magma code is recorded; no such original executable is supplied or priced as zero. The questions in Section 7 about other

binary markings, higher derived orthogonal groups and arbitrary-rank projective special linear constructions remain source-recorded questions, with no new answer or present-day open-status assertion. The corrected binary theorem, auxiliary counterexample and intermediate plus type are source corrections, not conjecture-credit claims.

References

[1] P. A. Brooksbank, J. T. Ferrara and D. Leemans, *Orthogonal groups in characteristic 2 acting on polytopes of high rank*, arXiv:1709.02219v2, supplied twelve-page edition.

[2] D. E. Taylor, *The Geometry of the Classical Groups*, Heldermann, 1992: Theorem 11.39 (larger-field symmetry generation), Corollary 11.42 (the binary exception), Theorem 11.43 (Dickson invariant). Author-hosted primary representation: `https://mail.maths.usyd.edu.au/u/don/papers/gcg.pdf`, printed pp.157–160, physical pp.165–168.

[3] D. Leemans, J. Moerenhout and E. O'Reilly-Regueiro, *Projective linear groups as automorphism groups of chiral polytopes*, supplied twenty-five-page edition, Proposition 2.3, p.5; its rank-one classification is explicitly borrowed from Bray–Holt–Roney-Dougal and Dickson, with external proof cost retained.

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge, 2002, Section 2E, in particular Proposition 2E16.